QM47400 · Spring 2027 · Final Project

Overview, Components and Calendar

EVENT	**Spring Undergraduate Research Conference, poster session, Tuesday, April 13, 2027.** Presenting is required for everyone
POSTER	Final poster (Milestone 10) due **Sunday, April 4, 2027, 11:59 PM ET**, nine days before the conference
WEIGHT	35% of the course grade. Poster-to-Product, after the conference, is a separate 5%
GROUPS	Groups of four, randomly assigned. One member uploads each group file, named with your group number (`NN`)

What the Final Project is

The Final Project is your group's capstone. You take a real prediction problem, build and evaluate models on real data, choose the one your stakeholder should use, and present the work as a research poster at Purdue's **Spring Undergraduate Research Conference** on Tuesday, April 13, with Professor Moreira as your faculty mentor. It runs as milestones M00 to M13 (no gaps in the numbering), plus a peer review, the conference and a peer evaluation.

The project must use **real data**: observations collected from actual people, organizations, transactions, markets, public records, surveys, experiments, devices or other real processes. Synthetic, simulated, artificially generated and AI-generated datasets are not eligible, and neither is the synthetic Bank Churn data of the Course Case Competition.

For inspiration, see award-winning posters from earlier editions of this course: https://davi-moreira.github.io/applied_projects.html. Conference information: https://www.purdue.edu/undergrad-research/conferences/index.php.

How it is graded

Component	Share of the project	Share of the course	What it covers
Milestone deliverables	30%	10.5%	The milestones below: clarity, completeness, timeliness
Peer evaluation (of teammates)	20%	7%	Your teammates' confidential ratings of you
Peer review (of posters)	10%	3.5%	Your group's feedback on other groups' posters
Conference presentation	20%	7%	Presenting your poster at the conference
Instructor/TA evaluation	20%	7%	Your final poster, scored with the poster rubric
Total	**100%**	**35%**	

Milestones and dates

Everything is due at 11:59 PM Eastern unless noted. Each item has its own PDF with the details.

Item	What you do	Opens	Due	Grading
M00	Round 01 meeting requests and group contract	Mon Feb 1	Sun Feb 7	completion
M01	Initial project proposal	Wed Jan 27	Sun Feb 7	50 points
M02	Expanded project outline	Mon Feb 8	Sun Feb 14	50 points
M03	Project draft abstract	Mon Feb 15	Sun Feb 21	30 points

Item	What you do	Opens	Due	Grading
M04	Round 01 meetings: confirmation	Thu Feb 18	Sun Feb 28	completion
M05	Apply to the conference (OURConnect)	Thu Feb 18	Sun Feb 28	completion
M06	Simple model and evaluation	Thu Feb 18	Sun Mar 7	100 points
M07	Round 02 meeting requests	Thu Feb 18	Sun Mar 7	completion
M08	More complex models and evaluation	Thu Feb 18	Sun Mar 14	100 points
M09	Poster first draft (`NN.pdf`)	Mon Mar 22	Sun Mar 28	poster rubric (feedback)
M10	Final poster (`NN.pdf`)	Mon Mar 29	Sun Apr 4	poster rubric
M11	Poster presentation planning	Sun Apr 4	Sun Apr 11	rubric
M12	Post an invitation to your presentation	Sun Apr 4	Sun Apr 11	completion
M13	Round 02 meetings: confirmation	Sun Apr 4	Sun Apr 11	completion
Peer review	Review five other groups' posters	Sun Apr 4	Sun Apr 11	10% of the project
Conference	Present your poster (required; in person)	Tue Apr 13	Tue Apr 13	20% of the project
P2P product	Poster-to-Product: team product	Fri Apr 16	Mon Apr 26	part of P2P's 5%
P2P live link	Poster-to-Product: your live link	Fri Apr 16	Wed Apr 28	part of P2P's 5%
Peer evaluation	Rate your teammates (individual)	Mon Apr 26	Fri Apr 30	20% of the project

The meeting cycle

In each of two rounds your group asks for a meeting with the TA and one with Professor Moreira, then confirms they happened. At least one member attends each meeting, not necessarily the same one, and there are no meetings during spring break (March 15 to March 20). Round 01 is about your prediction problem, your data and your modeling plan; Round 02 is about your models, your modeling strategy and your poster.

Round	Ask for the meetings	TA meeting	Meeting with Professor Moreira	Confirm
Round 01	M00, due Sun Feb 7	Mon Feb 8 to Sun Feb 21	Mon Feb 22 to Sun Feb 28	M04, due Sun Feb 28
Round 02	M07, due Sun Mar 7	Mon Mar 8 to Sun Mar 14, or Mon Mar 22 to Sun Mar 28	Mon Mar 29 to Sat Apr 3	M13, due Sun Apr 11

How your models are evaluated

As in the course from notebook 08 on: one 80/20 train/test split; all modeling on the training set with five-fold cross-validation; the test set opened once, at the end, for your chosen model. Models are compared on the same five folds through their per-fold differences, read against a margin δ set before comparing, as one of four outcomes (worthwhile gain, worthwhile loss, practically equivalent, unresolved). A more complex model replaces the baseline only on a worthwhile gain, so the baseline can be your champion. Milestones 06 and 08 walk you through it.

Good to know

- **The conference may not fall at our class time:** tell your other instructors early about any conflict. No class on Wednesday, April 14.
- **Peer review is not peer evaluation:** one is feedback on other groups' posters, the other your confidential rating of your teammates.
- **Honors contract?** The syllabus explains how your weights change. **Notebook checkpoints** ("M1" to "M4") are skill checks, not milestones.